

Analog Electronics - ED224

Course Placement: Analog Electronics is a core course for second year students of BTech EVD program.

Course format: It is 3 hours lecture and 3 hour laboratory every week.

Course content:

1. Introduction to Semiconductor Physics: N-type and P-type materials, electron and hole currents, PN junction and biasing
2. Diode Circuit Analysis: Diode model, Clipper and clamper circuits, Zener diodes and LED
3. Bipolar Junction Transistors: Transistor physics, NPN and PNP types, transistor ratings and specifications, transistor testing,
4. Transistor Amplifier Circuits: DC biasing, practical biasing circuits, BJT transistor amplifier configurations of common-emitter, -collector, and –base, equivalent circuit models, gain and impedance, troubleshooting common emitter amplifiers
5. Feedback amplifier: Designing of negative feedback amplifiers, voltage series, voltage shunt, current series and current shunt, stability analysis, unity-gain frequency of amplifier, phase-margin, compensation in multi-stage amplifiers
6. Differential amplifiers, Mono stable, Astable and bi- stable mutivibrators

Labs: Consists of hardware and software based practical on the above content.

Textbook:

Microelectronic Circuits by A. Sedra, K.C. Smith, 5th Edition. Oxford University Press.

Reference Book:

Analysis & Design of Analog Integrated Circuits by P. R. Gray, P. Hurst, S. Lewis, R. Meyer; John Wiley.

Grading Policy: 10 – 12 Laboratory Assignments, two mid-semester examinations and a final examination. 70 % weightage for written exam and 30 % for laboratory.

Course outcome: This course introduces the characteristics and applications of semiconductor devices and circuits. Emphasis is placed on analysis, selection, biasing, and applications. Upon completion, students should be able to construct, analyse, verify, and troubleshoot analog circuits using appropriate techniques and test equipment

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X	X			X	X	X	X

Lecture Schedule

Sl. No.	Description	No. of lectures
1	Introduction to Semiconductor Physics N-type and P-type materials, electron and hole currents, PN junction and biasing	1
2	Diodes	5
	2.1 Operation, Diode models,	2
	2.2 Rectifier circuits, Clipper and clamper circuits, voltage multiplier ckt., ...	2
	2.3 Zener diodes and LED	1
3	Bipolar Junction Transistors (BJTs)	29
	3.1 BJT Transistor operation and characteristics	2
	3.2 Biasing & bias stability of Transistors	1
	3.3 Small-signal analysis	4
	3.4 AC Equivalent circuit	2
	3.5 Capacitive coupling, emitter-bypassing	2
	3.6 Single-stage CE amplifiers: input/output impedance	4
	3.7 Emitter degeneration	2
	3.8 CB, CC amplifiers & applications	2
	3.9 Multi-stage amplifiers	1
	3.10 High-frequency transistor model	1
	3.11 Open-circuit time constant method for dominant pole	1
	3.12 Effect of decoupling capacitors: low-frequency cutoff	1
	3.13 Bode plot	1
	3.14 Basic concepts in Feedback	2
	3.15 Stability analysis	1
	3.16 Unity-gain frequency of amplifier, Phase-margin	1
	3.17 Compensation in multi-stage amplifiers: dominant pole compensation	1
4	Feedback amplifiers:	4
	4.1 Designing of negative feedback amplifiers, voltage series, voltage shunt,	2
	4.2 current series and current	2
5	Differential Amplifiers	2
6	Mutivibrators Astable, Monostable and Bistable Multivibrators	1